

Africast-Time Series Analysis & Forecasting Using R

<https://workshop.f4sg.org/africast/>

8. ARIMA models



Outline

- 1 ARIMA models
- 2 Seasonal ARIMA models
- 3 Forecast ensembles

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ARIMA models

- AR:** autoregressive (lagged observations as inputs)
- I:** integrated (differencing to make series stationary)
- MA:** moving average (lagged errors as inputs)

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An ARIMA model is rarely interpretable in terms of visible data structures like trend and seasonality. But it can capture a huge range of time series patterns.

Stationarity

Definition

If $\{y_t\}$ is a stationary time series, then for all s , the distribution of $(y_t, \dots, y_{t+s})$ does not depend on t .

Stationarity

Definition

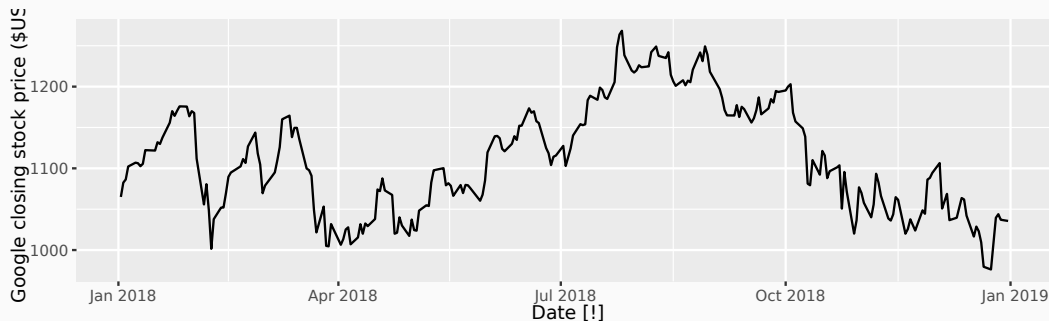
If $\{y_t\}$ is a stationary time series, then for all s , the distribution of $(y_t, \dots, y_{t+s})$ does not depend on t .

A **stationary series** is:

- roughly horizontal
- constant variance
- no patterns predictable in the long-term

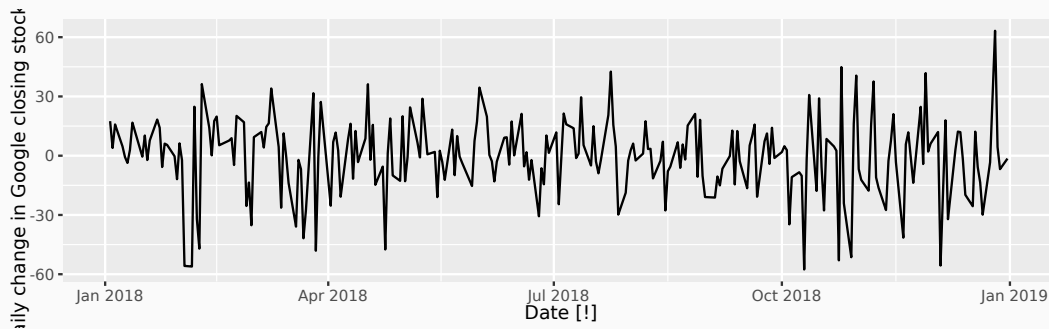
Stationary?

```
gafa_stock |>  
  filter(Symbol == "GOOG", year(Date) == 2018) |>  
  autoplot(Close) +  
  labs(y = "Google closing stock price ($US)")
```



Stationary?

```
gafa_stock |>  
  filter(Symbol == "GOOG", year(Date) == 2018) |>  
  autoplot(difference(Close)) +  
  labs(y = "Daily change in Google closing stock price")
```



Differencing

- Differencing helps to **stabilize the mean**.
- The differenced series is the *change* between each observation in the original series.
- Occasionally the differenced data will not appear stationary and it may be necessary to difference the data a second time.
- In practice, it is almost never necessary to go beyond second-order differences.

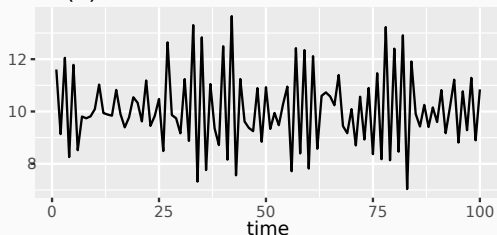
Autoregressive models

Autoregressive (AR) models:

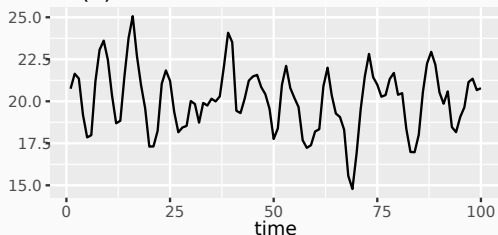
$$y_t = c + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \dots + \phi_p y_{t-p} + \varepsilon_t,$$

where ε_t is white noise. A multiple regression with **lagged values** of y_t as predictors.

AR(1)



AR(2)



- Cyclic behaviour is possible when $p \geq 2$.

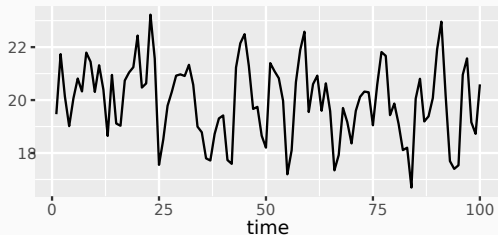
Moving Average (MA) models

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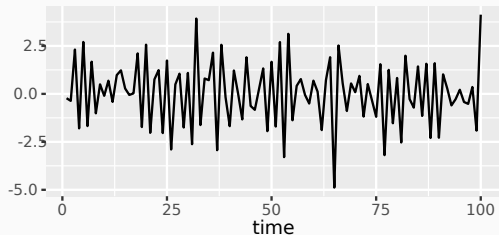
$$y_t = c + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \dots + \theta_q \varepsilon_{t-q},$$

where ε_t is white noise. A multiple regression with **lagged errors** as predictors. *Don't confuse with moving average smoothing!*

MA(1)



MA(2)



ARIMA models

Autoregressive Moving Average models:

$$y_t = c + \phi_1 y_{t-1} + \dots + \phi_p y_{t-p} \\ + \theta_1 \varepsilon_{t-1} + \dots + \theta_q \varepsilon_{t-q} + \varepsilon_t.$$

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- Predictors include both **lagged values of y_t** and **lagged errors**.

ARIMA models

Autoregressive Moving Average models:

$$y_t = c + \phi_1 y_{t-1} + \dots + \phi_p y_{t-p} \\ + \theta_1 \varepsilon_{t-1} + \dots + \theta_q \varepsilon_{t-q} + \varepsilon_t.$$

- Predictors include both **lagged values of y_t** and **lagged errors**.

Autoregressive Integrated Moving Average models

- Combine ARMA model with **differencing**.
- d -differenced series follows an ARMA model.
- Need to choose p, d, q and whether or not to include c .

ARIMA models

ARIMA(p, d, q) model

AR: p = order of the autoregressive part

I: d = degree of first differencing involved

MA: q = order of the moving average part.

- White noise model: ARIMA(0,0,0)
- Random walk: ARIMA(0,1,0) with no constant
- Random walk with drift: ARIMA(0,1,0) with const.
- AR(p): ARIMA($p,0,0$)
- MA(q): ARIMA(0,0, q)

Example: National populations

```
fit <- global_economy |>  
  model(arima = ARIMA(Population))  
fit
```

```
# A mable: 263 x 2
```

```
# Key:      Country [263]
```

Country	arima
<fct>	<model>
1 Afghanistan	<ARIMA(4,2,1)>
2 Albania	<ARIMA(0,2,2)>
3 Algeria	<ARIMA(2,2,2)>
4 American Samoa	<ARIMA(2,2,2)>
5 Andorra	<ARIMA(2,1,2) w/ drift>
6 Angola	<ARIMA(4,2,1)>
7 Antigua and Barbuda	<ARIMA(2,1,2) w/ drift>
8 Arab World	<ARIMA(0,2,1)>
9 Argentina	<ARIMA(2,2,2)>

Example: National populations

```
fit |>  
  filter(Country == "Australia") |>  
  report()
```

Series: Population

Model: ARIMA(0,2,1)

Coefficients:

ma1

-0.661

s.e. 0.107

sigma^2 estimated as 4.063e+09: log likelihood=-699

AIC=1401 AICc=1402 BIC=1405

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Series: Population

Model: ARIMA(0,2,1)

Coefficients:

ma1
-0.661
s.e. 0.107

$$y_t = 2y_{t-1} - y_{t-2} - 0.7\varepsilon_{t-1} + \varepsilon_t$$
$$\varepsilon_t \sim \text{NID}(0, 4 \times 10^9)$$

sigma^2 estimated as 4.063e+09: log likelihood=-699

AIC=1401 AICc=1402 BIC=1405

Understanding ARIMA models

- If $c = 0$ and $d = 0$, the long-term forecasts will go to zero.
- If $c = 0$ and $d = 1$, the long-term forecasts will go to a non-zero constant.
- If $c = 0$ and $d = 2$, the long-term forecasts will follow a straight line.
- If $c \neq 0$ and $d = 0$, the long-term forecasts will go to the mean of the data.
- If $c \neq 0$ and $d = 1$, the long-term forecasts will follow a straight line.
- If $c \neq 0$ and $d = 2$, the long-term forecasts will follow a quadratic trend.

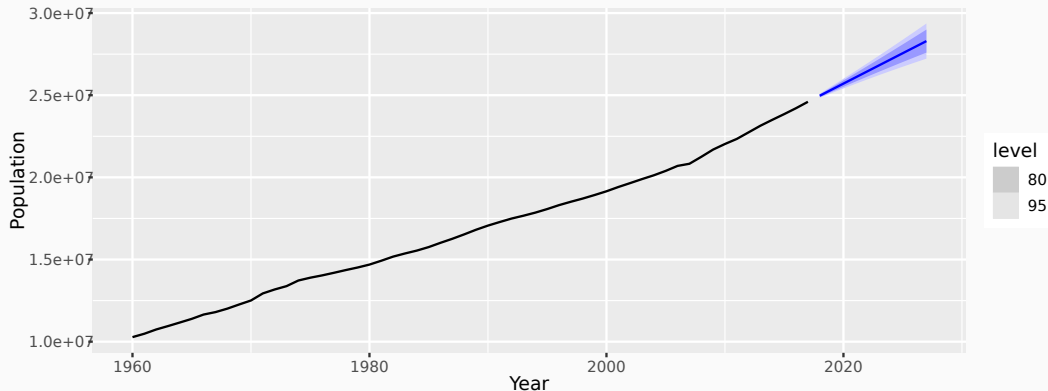
Understanding ARIMA models

Forecast variance and d

- The higher the value of d , the more rapidly the prediction intervals increase in size.
- For $d = 0$, the long-term forecast standard deviation will go to the standard deviation of the historical data.

Example: National populations

```
fit |>  
  forecast(h = 10) |>  
  filter(Country == "Australia") |>  
  autoplot(global_economy)
```



How does ARIMA() work?

Hyndman and Khandakar (JSS, 2008) algorithm:

- Select no. differences d via KPSS test.
- Select p, q and inclusion of c by minimising AICc.
- Use stepwise search to traverse model space.

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$$\text{AICc} = -2 \log(L) + 2(p + q + k + 1) \left[1 + \frac{(p + q + k + 2)}{T - p - q - k - 2} \right]$$

where L is the maximised likelihood fitted to the *differenced* data, $k = 1$ if $c \neq 0$ and $k = 0$ otherwise.

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where L is the maximised likelihood fitted to the *differenced* data, $k = 1$ if $c \neq 0$ and $k = 0$ otherwise.

Note: Can't compare AICc for different values of d .

How does ARIMA() work?

Step1: Select current model (with smallest AICc) from:

ARIMA(2, d , 2)

ARIMA(0, d , 0)

ARIMA(1, d , 0)

ARIMA(0, d , 1)

How does ARIMA() work?

Step1: Select current model (with smallest AICc) from:

ARIMA(2, d , 2)

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ARIMA(0, d , 1)

Step 2: Consider variations of current model:

- vary one of p, q , from current model by ± 1 ;
- p, q both vary from current model by ± 1 ;
- Include/exclude c from current model.

Model with lowest AICc becomes current model.

How does ARIMA() work?

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ARIMA(2, d , 2)

ARIMA(0, d , 0)

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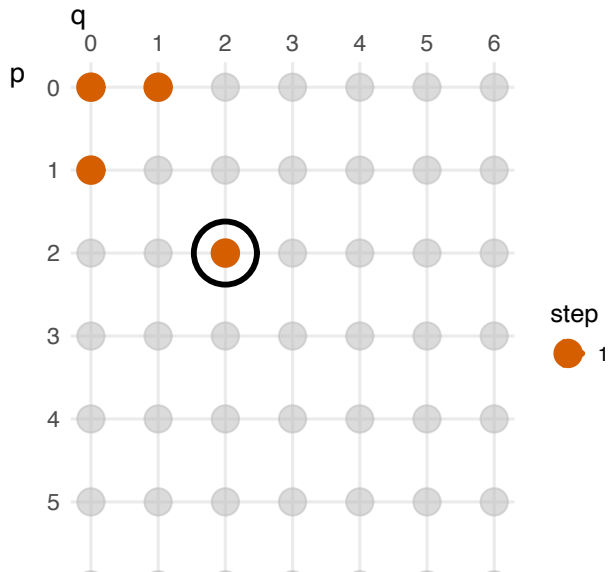
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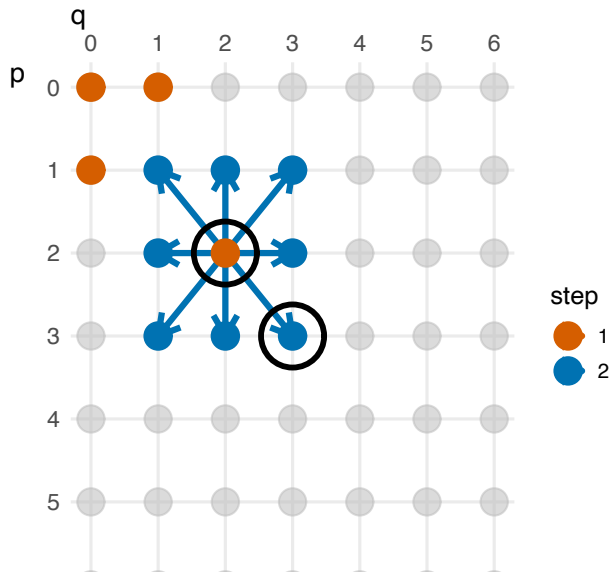
Model with lowest AICc becomes current model.

Repeat Step 2 until no lower AICc can be found.

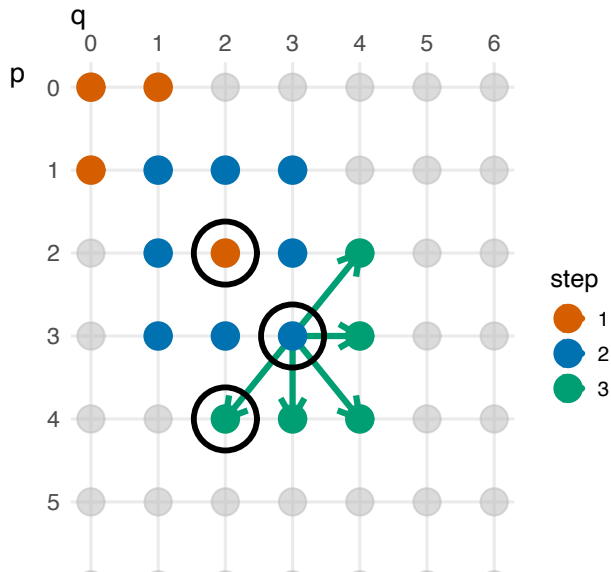
How does ARIMA() work?



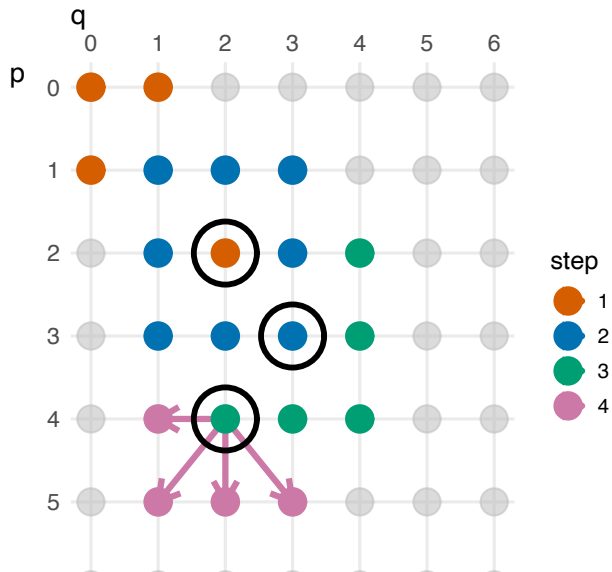
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How does ARIMA() work?



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Seasonal ARIMA models

ARIMA	(p, d, q)	$(P, D, Q)_m$
	↑	↑
	Non-seasonal part of the model	Seasonal part of of the model

- m = number of observations per year.
- d first differences, D seasonal differences
- p AR lags, q MA lags
- P seasonal AR lags, Q seasonal MA lags

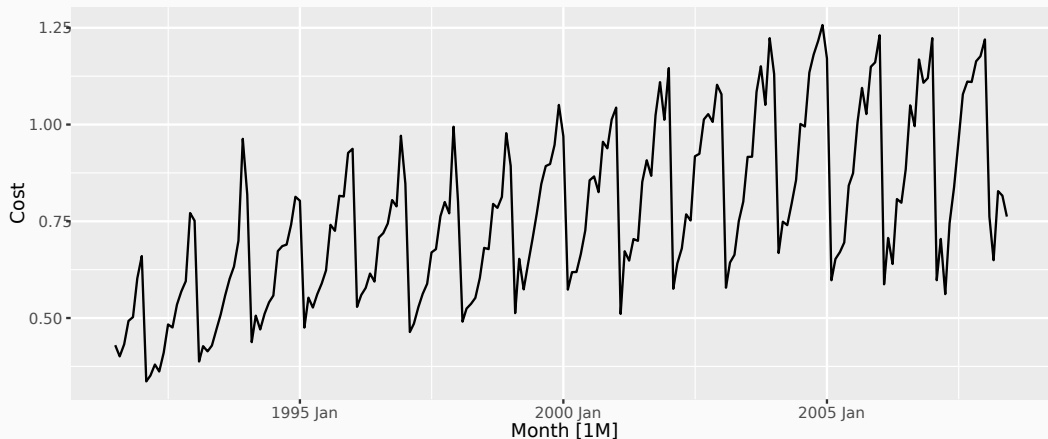
Seasonal and non-seasonal terms combine multiplicatively

Corticosteroid drug sales

```
h02 <- PBS |>  
  filter(ATC2 == "H02") |>  
  summarise(Cost = sum(Cost) / 1e6)
```

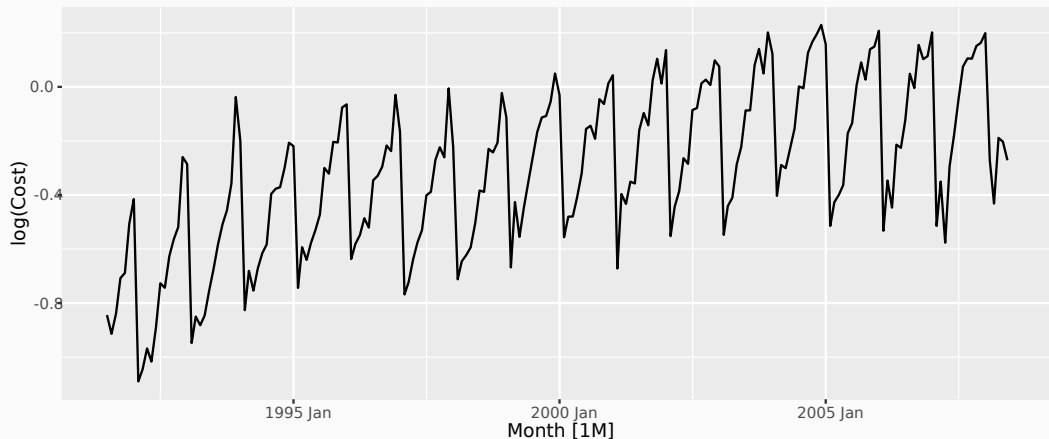
Corticosteroid drug sales

```
h02 |> autoplot(  
  Cost  
)
```



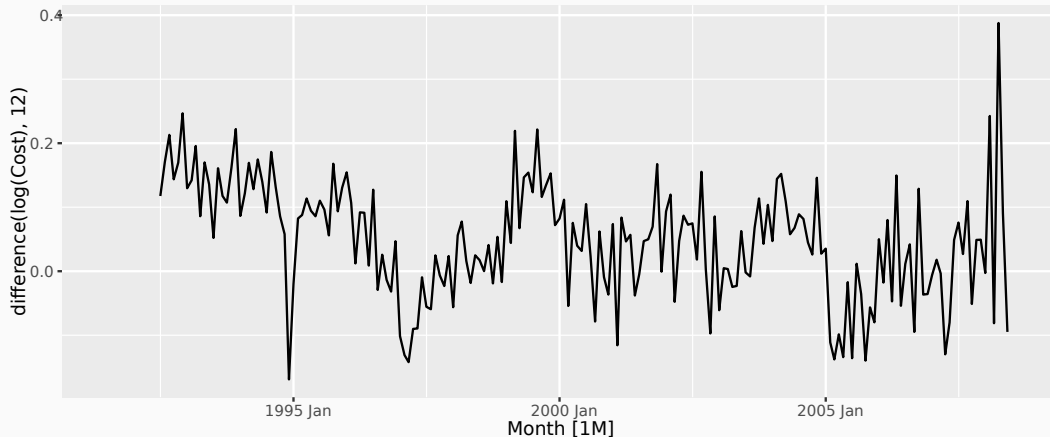
Corticosteroid drug sales

```
h02 |> autoplot(  
  log(Cost)  
)
```



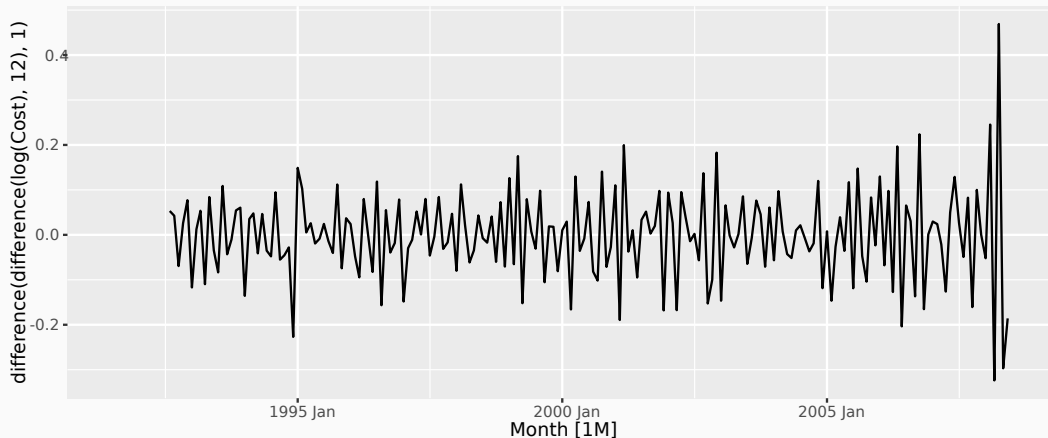
Corticosteroid drug sales

```
h02 |> autoplot(  
  log(Cost) |> difference(12)  
)
```



Corticosteroid drug sales

```
h02 |> autoplot(  
  log(Cost) |> difference(12) |> difference(1)  
)
```



Corticosteroid drug sales

```
h02 |>  
  model(arima = ARIMA(log(Cost))) |>  
  report()
```

Series: Cost

Model: ARIMA(2,1,0)(0,1,1)[12]

Transformation: log(Cost)

Coefficients:

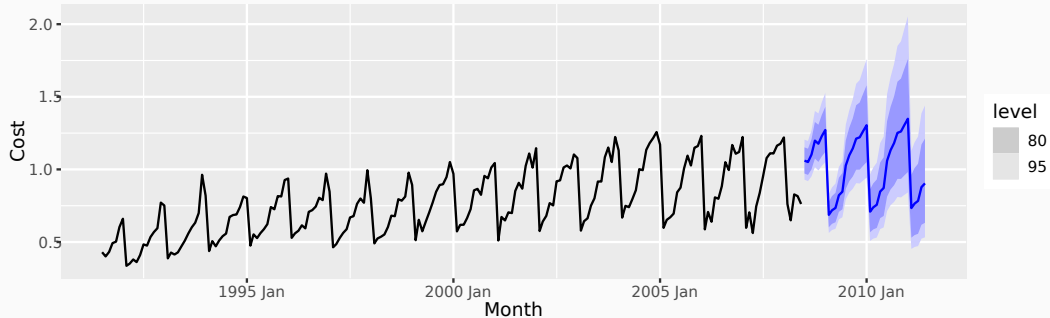
	ar1	ar2	sma1
	-0.8491	-0.4207	-0.6401
s.e.	0.0712	0.0714	0.0694

sigma^2 estimated as 0.004387: log likelihood=245

AIC=-483 AICc=-483 BIC=-470

Corticosteroid drug sales

```
h02 |>  
  model(arima = ARIMA(log(Cost))) |>  
  forecast(h = "3 years") |>  
  autoplot(h02)
```



Corticosteroid drug sales

```
fit <- h02 |>
  model(best = ARIMA(log(Cost),
    stepwise = FALSE,
    approximation = FALSE,
    order_constraint = p + q + P + Q <= 9
  ))
report(fit)
```

Series: Cost

Model: ARIMA(4,1,1)(2,1,2)[12]

Transformation: log(Cost)

Coefficients:

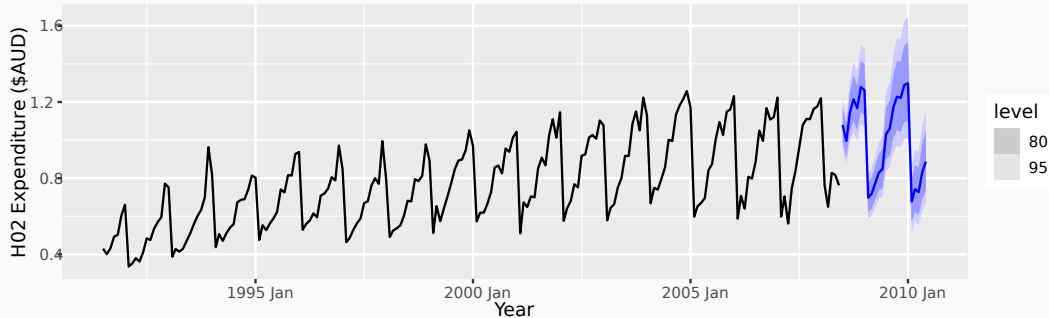
	ar1	ar2	ar3	ar4	ma1	sar1	sar2	sma1	sma2
	-0.0425	0.210	0.202	-0.227	-0.742	0.621	-0.383	-1.202	0.496
s.e.	0.2167	0.181	0.114	0.081	0.207	0.242	0.118	0.249	0.213

sigma^2 estimated as 0.004049: log likelihood=254

AIC=-489 AICc=-487 BIC=-456

Corticosteroid drug sales

```
fit |>  
  forecast() |>  
  autoplot(h02) +  
  labs(y = "H02 Expenditure ($AUD)", x = "Year")
```



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Forecast ensembles

```
train <- tourism |>
  filter(year(Quarter) <= 2014)
fit <- train |>
  model(
    ets = ETS(Trips),
    arima = ARIMA(Trips),
    snaive = SNAIVE(Trips)
  ) |>
  mutate(mixed = (ets + arima + snaive) / 3)
```

- Ensemble forecast `mixed` is a simple average of the three fitted models.
- `forecast()` will produce distributional forecasts taking into account the correlations between the forecast errors of the component models.

Forecast ensembles

```
fc <- fit |> forecast(h = "3 years")  
fc |>  
  filter(Region == "Snowy Mountains", Purpose == "Holiday") |>  
  autoplot(tourism, level = NULL)
```

